Games in Africa: Opportunities & Challenges



Growth of mobile games in Africa is a way to retell local legends and stories and portray them in a modern, appealing way

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[1] *Owari a traditional game played in West Africa*
Image source: http://en.wikipedia.org/wiki/Oware#mediaviewer/File:Owari-front.jpg *CC BY-SA 3.0*

Table of Contents

I. Introduction 3

II. Community 3

III. Platforms (device & connectivity) & Market 5

IV. Type of games & Target Groups 10

V. Business models 12

Game Business models 12

Startup business models 13

VI. Challenges 14

Payments 14

Visibility & discoverability 15

Access to finance 16

Access to capacities 16

VII. Conclusion 17

VIII. Acknowledgements 18

I. Introduction

Mobile phone ownership and use is on a meteoric rise all across Africa and the inevitable rise in mobile gaming. According to a recent study by eMarketer[2], mobile phone usage across Africa and the Middle East nearly doubled in 2013, and smartphones now make up more than 20% of all mobile phones being used. The study estimated around 525.8 million people of all ages used a mobile phone at least every month during 2013. There is an estimated 561-million mobile phone users and that number will grow to 670.9 million by 2017.

GSMA in its 2013 study of the state of mobile economy in Sub-Saharan Africa[3] reports that Sub-Saharan Africa's unique mobile subscriber base has grown by 18 per cent annually over the last five years, making it the fastest growing region globally. By mid-2013, there were 253 million unique mobile subscribers and 502 million connections. With many countries in the region seeing fixed line penetration rates of less than five per cent, mobile has emerged as the main medium for accessing the internet across Sub-Saharan Africa. While 2G connections still dominate in the region, 3G and 4G networks are gaining scale and smartphone ownership is on the rise. With unique subscriber penetration rates still less than 33 per cent, this opens up a major opportunity for growth in the next five years.

The numbers show how lucrative the mobile market is in Africa. What kind of open doors does this growing market open for game and mobile app developers? With smartphones penetrating the African market at such an exponential rate, what is the current state of mobile gaming in Africa? This study realized for Orange aims to address this exciting potential and this document is a summary of findings based on desk research and interviews.

The document is split in five subsections:

- The first subsection presents the state of the game developers' communities in Sub-Saharan Africa.
- The second subsection focuses on the gaming platform
- The third subsection presents the findings related to the types of games and the target groups
- The fourth subsection talks about business models for both games and game studios.
- Finally, the last part underlines the various challenges that have been identified during the study

II. Community

This section presents the state of the game developers' communities in Sub-Saharan Africa as well as the state of game studios in various countries. The findings are based on interviews and desk research conducted in various countries including Senegal, Ghana,

[2] Source: http://www.emarketer.com/Article/Smartphone-Usage-Nearly-Double-Middle-East-Africa/1010249

[3] Source: http://www.gsmamobileeconomyafrica.com/Sub-Saharan%20Africa_ME_Report_English_2013.pdf

Kenya, Uganda, South-Africa, Nigeria and Cameroon. In order to have a sense of the state in these countries, interviews were conducted with managers of technology hubs, managers of technology incubators, game developers, managers of game communities and manager of game studios. See the acknowledgements section to get an exhaustive list of interviewees.

Globally, in all the countries covered in the study, the community of game developers exists and is structured. Almost all tech hubs have sub-communities interested in games, and there are dedicated events focused on game development. In some countries, the community is informal, in some other countries it is more formalized either as an association (e.g. South Africa and Make Game South-Africa[4]) or as a chapter of international associations like the International Game Developers Association (IGDA)[5]. IGDA has a few local chapters in Africa, and is organizing regular Games Camps in various countries. See figure 1 for details.

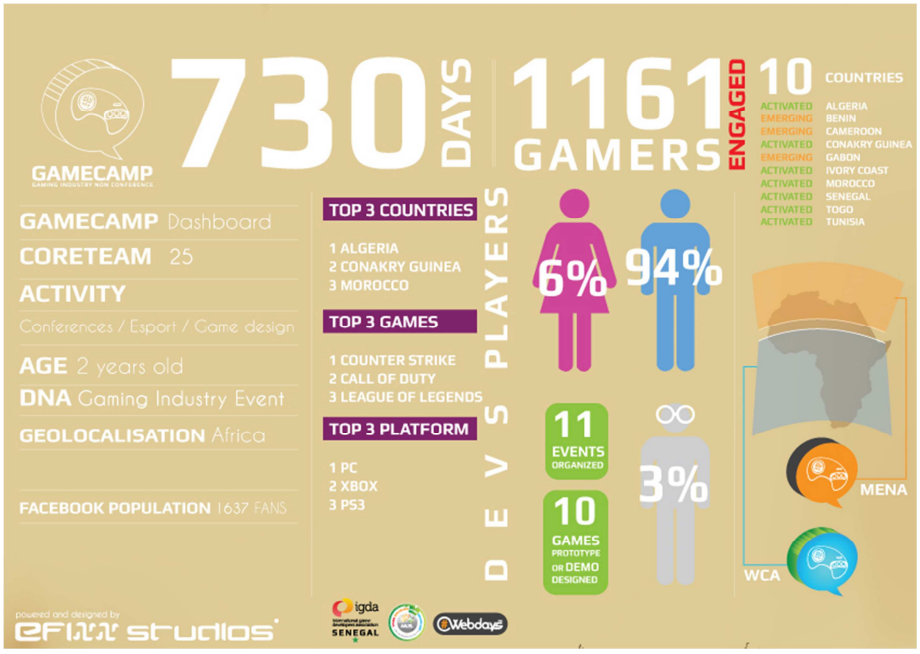


Figure 1: IGDA Games Camp in Africa

In the same way, there are other game events being organized from time to time in various countries (e.g. the World Day of Video Games in Dakar[6]) that highlight the interests from local gamers as well the existence of a potential market.

Another evidence of the growth of the Game development communities is the number of games submitted to various application competitions like Pivot East[7] in Kenya or Tekki48[8] in Senegal. The Tekki48 2014 winner in Senegal was a game company (Cauriolis[9]) and Pivot East has a dedicated section to Games and Entertainment, and a Game won the overall competition in 2012. All these elements highlight the interest from developers to focus on game developments, and also show the opportunities offered by local markets.

[4] http://makegamessa.com/
[5] http://www.igda.org/
[6] https://www.facebook.com/JMJVSenegal
[7] http://www.pivoteast.com/
[8] https://fr-fr.facebook.com/Tekki48
[9] https://fr-fr.facebook.com/cauriolis

In terms of game start-ups, the situation is very different and countries are at very different stages. The industry is flourishing in South-Africa (32 game studios today), Kenya and Nigeria. In Kenya, a few game startups (from Kenya but also from neighboring countries like Uganda) have been incubated and have launched various games. In Nigeria, there are numerous studios developing a number of popular casual games. In these three countries, game is a real active subsector. However, it is important to note that the take-off of the sector in these countries is very recent (2 years or so). The situation is very different in other countries, particularly in Francophone countries. The conditions for the market development are present: there are a number of developers aware of and excited by game development, the critical mass is here in terms of market (players with smartphones), but the number of companies in the sector is very limited. This is largely due to a few issues that are described in the challenges section. In this context, a small set of specific interventions may be change the sector dramatically.

III. Platforms (device & connectivity) & Market

Concerning the preferred game platforms, the study focused on platforms available in the local market. A few games and game studios, particularly in South-Africa, are targeting the international market on PC and Consoles. These platforms are not the most widely available to people, and the study specifically focused on the mobile platform.

As a starting point, it is important to note the importance of games on average mobile usage in Africa. A study by Ericsson in 2013[10] showed that gaming is one of the core activities on mobile phones. The figure below shows the mobile feature used weekly by users in Africa.

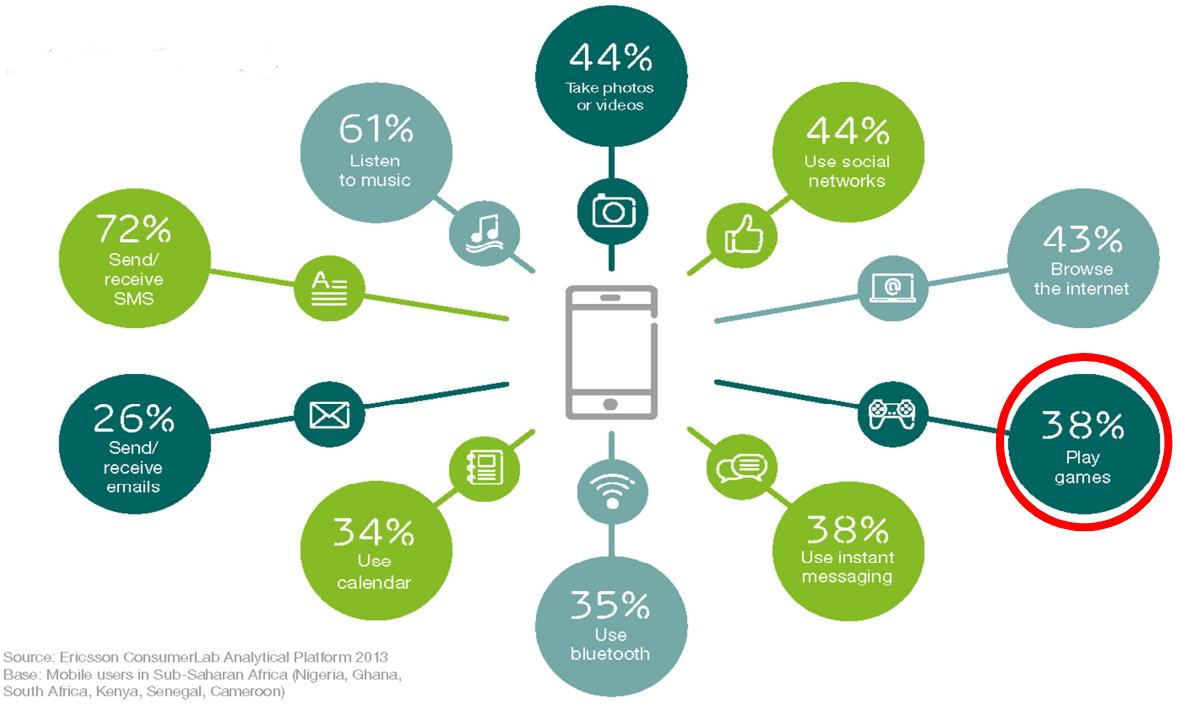


Figure 2: Top 10 mobile features used weekly in Sub-Saharan Africa

[10] http://www.ericsson.com/res/docs/2013/consumerlab/bridging-the-digital-divide-sub-saharan-africa.pdf

However, it is important to have a finer grain analysis on the population and type of phones. On the mobile platform, the installed base can be split in 3 segments:

- Basic phones (phone calls+SMS and USSD)
- Feature phones (J2ME)
- Smartphones (Android/Windows/IoS)

Basic phones are not really gaming platforms. In this category only SMS-based games are available. As per the research, only lottery/betting games are using SMS and premium rate. Feature phones and smartphone are the two platforms that have real capabilities for game development.
As of today, it is clear that the mass-market is on feature phones. A study by VisionMobile in 2012[11] showed that only 27% of the connected phones are smartphone, 73% of the phones accessing the internet in Africa are feature phones. See Figure 3 below.

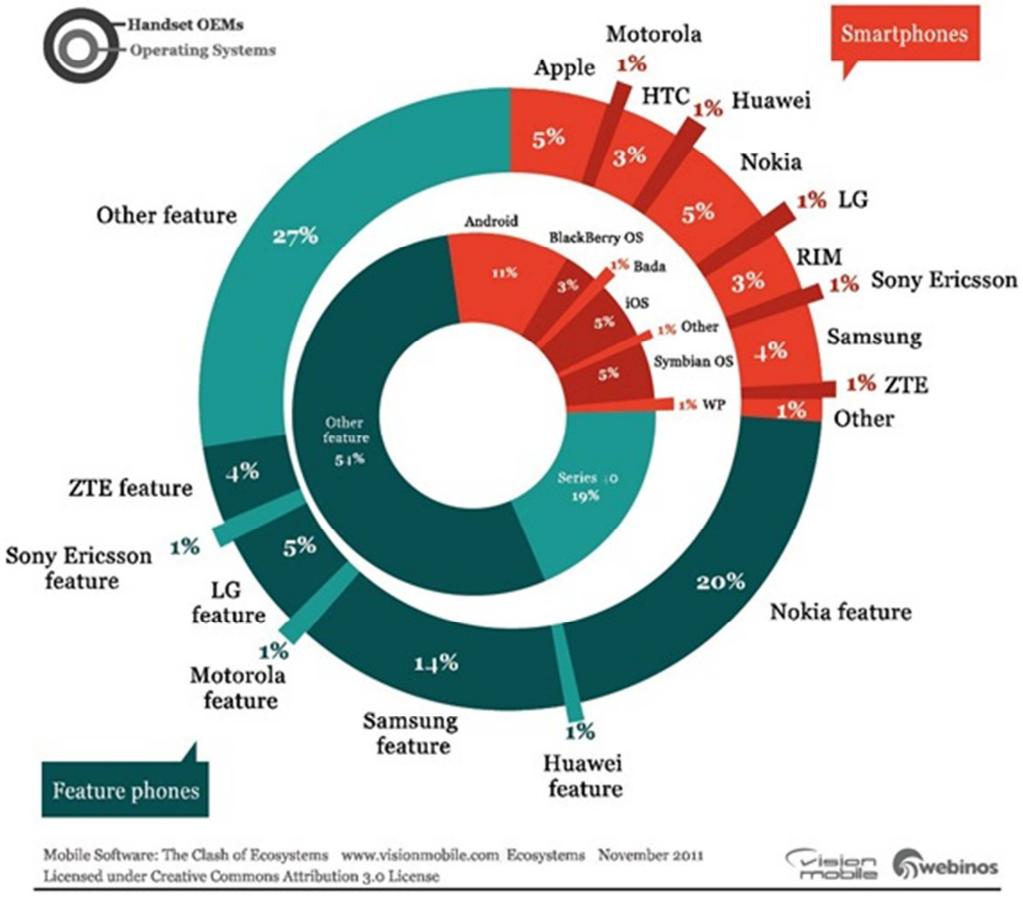
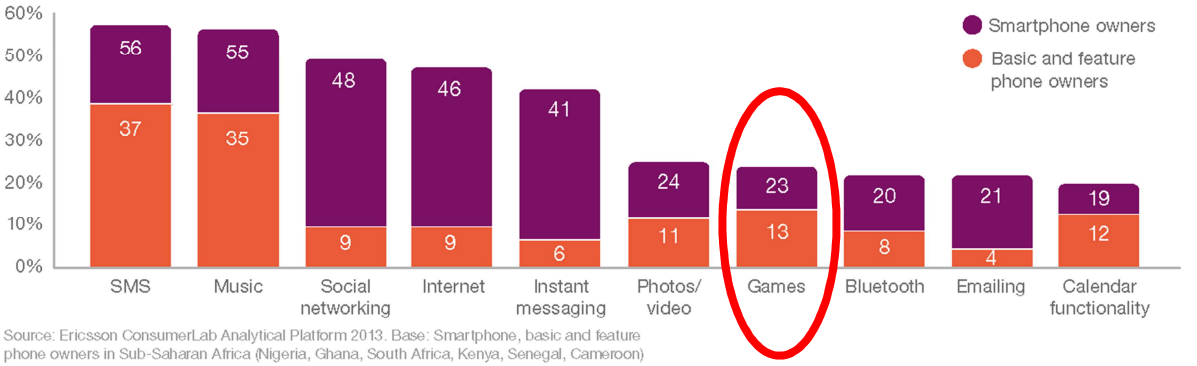


Figure 3: market repartition in 2012 of connected phones

However, another statistic from the Ericsson's study is important to take into account concerning activities per type of mobile platforms. The figure 4 below shows that for gaming, the proportion of people playing game is higher on smartphone than on feature phones.

[11] http://www.bandwidthblog.com/2012/02/22/why-you-shouldnt-forget-about-the-feature-phone-in-africa/

Figure 1: Mobile features used on a daily basis



Source: Ericsson ConsumerLab Analytical Platform 2013. Base: Smartphone, basic and feature phone owners in Sub-Saharan Africa (Nigeria, Ghana, South Africa, Kenya, Senegal, Cameroon)

Figure 4: Mobile features used on a daily basis in Sub-Saharan Africa

It is also clear that the population that is more likely playing games (teenagers and young adults) is usually more tech-savy and has more advanced phones. It is to be noted that this was qualitatively mentioned by almost all interviewees, but we don't have specific statistics on smartphone penetration in the 16-35 groups that could highlight (or deny) this fact.

Apart from the penetration rate, two other key dimensions for the discussion on J2ME versus smartphone are related to specific challenges of development and deployment. On development, the interoperability of various versions and implementations of J2ME is a major problem for portability of games across various handsets. This was largely highlighted by many developers the study interviewed. The second issue is the distribution platform. With Nokia announcing the closing of OVI Store in January 2014, there are now no more easy options for developers to make their games available to users. This is clearly a major push-back for the developers who put their faith and technical resources on such platforms. Based on these elements, we believe that the J2ME platform is not the right platform to focus on for the future, and its importance is very likely going to decrease quickly, at least in the target group for games.

Now, looking into the finer details in the smartphone category, it is clear that the Android platform is by far the most popular operating system. The figure 5 below shows the respective market share of various mobile operating systems. The importance of Android has been largely underlined by all the interviewees. However, it is also important to note that a big part of the interviewees also mentioned their interest in and the development on the Windows OS platform. While the market share of Windows OS phone is still very low and not yet taking off (figure 5), the situation may change quickly with the recent announce of Nokia and Microsoft. Microsoft will now provide windows os for free, and it will be the core platform for nokia entry-level smartphones. Given the importance of the Nokia brand in Africa (see figure 6), this is likely going to have an impact on the number of windows os phone on the market. Finally, while there were major media announces recently concerning firefox OS on low-cost smartphones targeting developing regions, none of the interviewees mentioned this operating system. In the same way, while the mobile web seems to an attractive platform (instead of native applications) for developers (see figure 7), only one of the interviewees (Maliyo Games) mentioned it as a platform for game development.

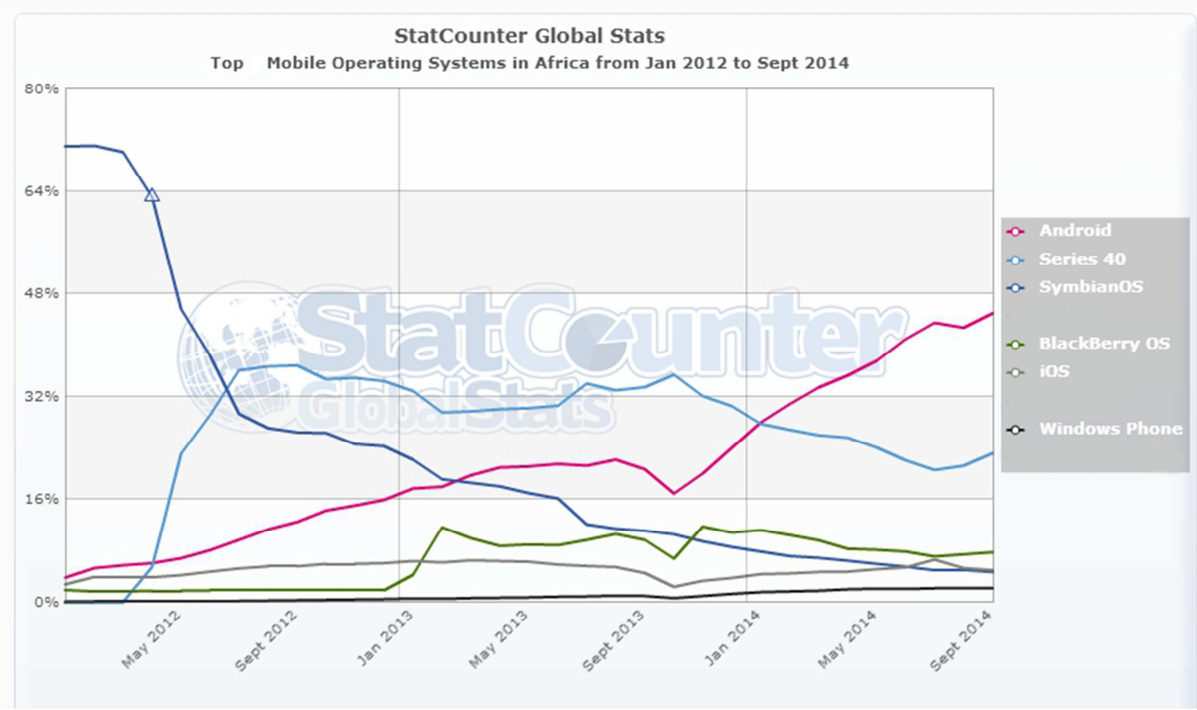


Figure 5: Top mobile operating systems from Jan 2012 till Sept 2014

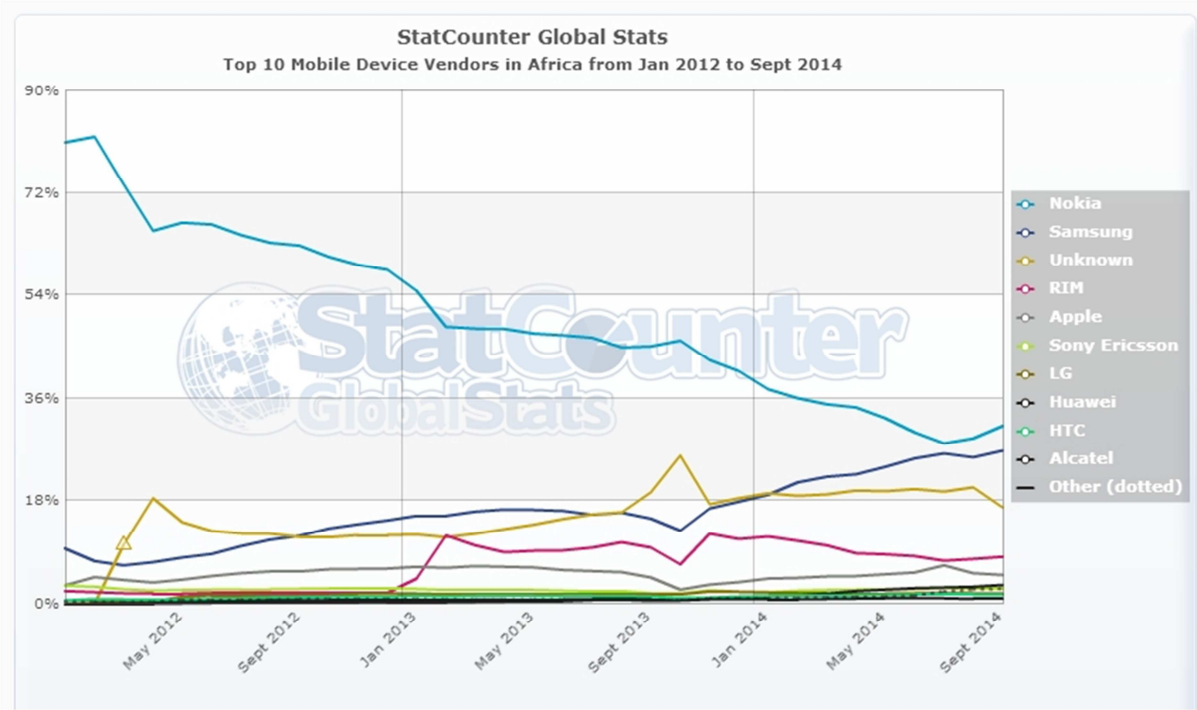


Figure 6: Top mobile device vendors in Africa

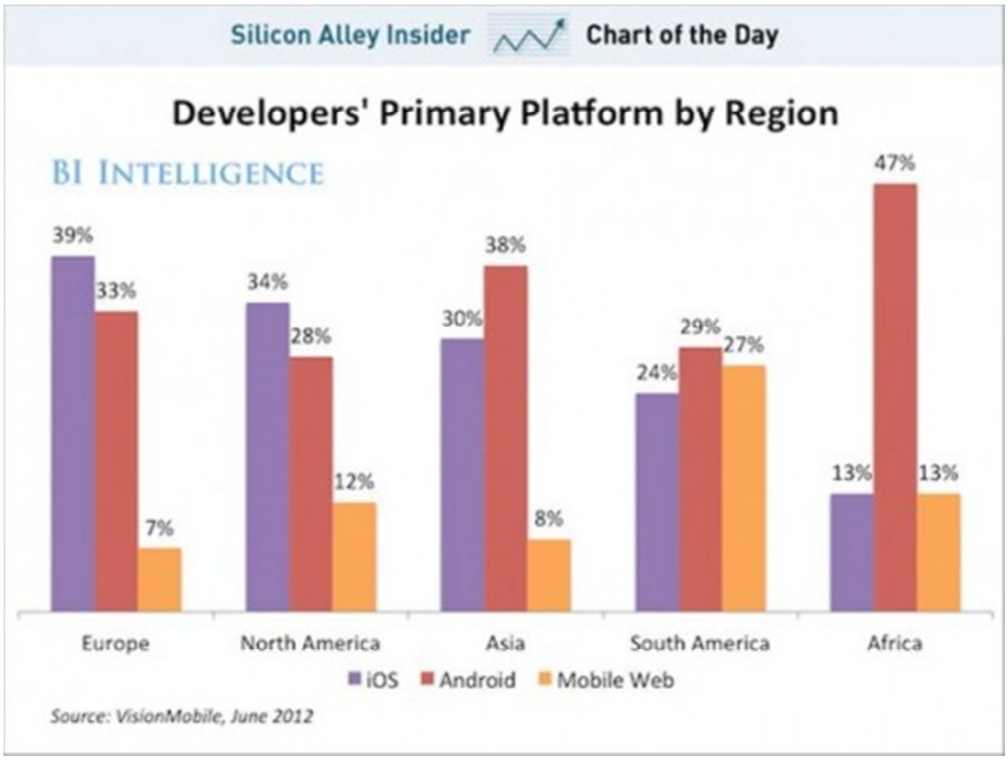


Figure 7: Smartphone OS market share in various region of the world

In a similar way, a study by the World Bank[12] in Ghana and Nigeria showed the same kinds of statistics. See figure 8 below.

		Ghana	Nigeria
Responses		800	800
Female		23.3%	30.3%
Average age		24	23
Minimum age		15	15
Maximum age		61	65
Rural responses		31.8%	24.3%
Installed mobile apps		93.8%	92.9%
App stores previously used	Google Play	60.8%	52.4%
	Nokia Ovi	29.6%	39.5%
	MTN app Store	20.8%	23.1%
	Samsung app store	15.5%	6.5%

(*Source:* Jana mobile user survey)

Figure 8: Survey of Smart and Feature Phone Users in Nigeria and Ghana

[12] http://www.infodev.org/infodev-files/mobile_apps_at_the_base_of_the_pyramid_summary_report.pdf

Interviewees highlighted the fact that not only the development on iOS is more expensive, as the developer needs a Mac and an iPhone, but the publication on Apple store is far more difficult than on Android or Windows store. The use of stores is essential as it is the primary way for users to download application. Users very rarely install other application stores than the one already installed on the handset, and are also very rarely downloading applications directly outside the stores.

IV. Type of games & Target Groups

The section covers the type of games that are being developed and the specific target groups.

Concerning target groups, various indicators provided by desk research as well as feedback from interviewees highlighted two characteristics of the typical gamers:

- The typical gamer is a teenager or young adult (<35)
- The proportion of Male gamers is much higher compared to Female gamers. This appears clearly in Figure 1 with the female attendance to games camp at 6%. It also appears clearly in figure 8 with the proportion of Female representation <25%.

These characteristics apply to entertainment-focused games and not serious games or education games. For those games, to the best of our research, there are not specific studies available that would provide disaggregated data on the target groups.

Concerning types of game, based on the interviews, it makes sense to consider two categories:

- Serious games
- Entertainment games

Concerning serious games, the vast majority of products are focused not on education but on social issues, particularly related to health and sanitation. Games about HIV protection, malaria protection, and related issues are the ones that were cited most by developers. On education, examples include products that are not really games but are using gamification approaches, like quizzes etc. It is important to note that there is a growing interest from public and private donors to invest in gaming. While there is no formal research available highlighting the impact and education outcomes of such games, there is more investment across various domains (education, agriculture, health, social issues etc.) by several donors. Few international initiatives now have dedicated section on gaming approach. An example is the mEducation Alliance[13] that has had various seminars on the topic, and is regularly highlighting games at its annual symposium (see e.g. 2013 mEducation Alliance Symposium[14] highlighting multiple games or software using gamification approaches).

[13] http://www.meducationalliance.org/event/seminar-series-december-2010-mobiles-games-literacy

[14] http://www.meducationalliance.org/document/projects-presented-2013-meducation-alliance-symposium

Concerning entertainment games, there are clearly two categories. Some studios focus on the international market owing to the challenges of addressing local markets (see Challenges section below). They see more opportunities for revenue due to higher purchasing-power for user in the north and are therefore developing and designing games that will compete at the international level. An example of such a game is iWarrior by Leti Games with 8000+ download, mostly in the US, 3000+ paid users (75% in the US).

There is a second category of studios that are targeting the local market and investing a lot in the localization of the game. The approach is well summarized in the figure 9 below from the studio Kiro'o Games[15] in Cameroon.

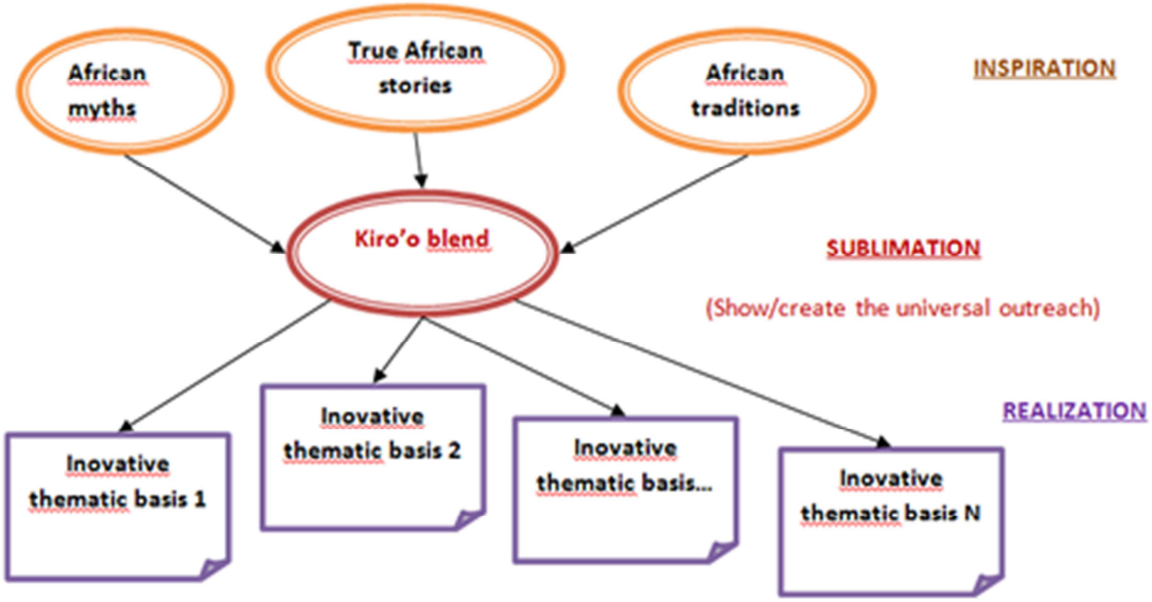


Figure 9: Kiro'o Games Approach to game development

This approach is aiming at providing meaningful and exciting games to the local markets. There are a growing number of studios taking this approach and even having a country approach. A good example is Leti Games that released Spar, a very popular Ghanaian card games. They also released Matatu, a very popular card games in Uganda. The results are not significantly different but the market is local. For example Leti's Matatu got 6850+ downloads, more than 20.000 games played every day, and 75% of the players are in Uganda.

Finally, it is important to note that at this point in time, all mobile games developed for the local markets are mono-player games, not using any specificity of the mobile phones such as geo-location. There is currently no real demand from games developers to have framework for multi-player games or access to advanced technical enablers such as device positioning. This is largely due to the market not being very developed yet, but it is likely that innovations will take place in the future if the opportunities exist, and if competition arises. That said, it is important to note that a few interviewees mentioned the need for a technical platform to support a specific payment scheme based on premium rate SMS and codes. In order to solve the payment challenges, a few developers

[15] http://kiroogames.com/en/kiro-o-tales.html

mentioned the development of an hybrid solution based on in-app payment through codes obtained by SMS using a premium rate number. Given that a few developers had the exact same idea (generic functionalities), and all mentioned the complexity of developing such a platform, the development and release of such a platform might be a powerful and easy-to-do technical and business enabler..

V. Business models

The section describes the various business models we identified during the desk research and the interviews. The section has two parts discussing the business models of games, and the business models of game studios.

Game Business models

There are four clear options for game developers to generate revenues from a game:

- **Pay-per-download (aka premium model)**: Gamers pay for the game before they can download it
- **In-app payments (aka freemium model)**: Gamers, during the game, can buy extra functionalities, extensions or helpers
- **Advertisement**: During the game, gamers watch various advertisements, and the advertiser pays the game developers
- **Subscription**: Gamers pay for a daily, weekly or monthly subscription.

In the market at large, these models are not equally represented. The pay-per-download model is clearly getting less traction all over the world. In Sub-Saharan Africa, due to issues related to payments, plus piracy, this model does not seem to be a viable option to most of the interviewees.

The subscription model is not getting traction at the individual game level. We didn't find any example of a game based on such a model. However, this model is very popular on game portals, especially mobile operator game portals like Safaricom Games Lounge or Airtel Games club Gamers pay a subscription to access a complete portfolio of games, usually well-known branded ones like Electronic Arts blockbusters (Fifa series, etc.).

The advertisement model has been by far the most used one. However, generating revenue in this model is not easy. Indeed, a series of factors are impacting this model. This includes:

- **Required expertise in the various advertisement models**: Some ad-networks pay based on the size of the ad, some pay on the number of clicks, some pay on the length (in second) the ad is visible etc. The selection of the ad-network and model has to happen during the design time to maximize the revenue
- **Required expertise in user experience**: Integrating advertisements in app is a very difficult process. It is essential to find the right balance between revenues and customers' unhappiness due to ads. Nowadays, customers in general are less and less ready to accept intrusive advertisements, and they are more likely going to drop a game if they are bored by the level of advertisements.

- **Popularity**: Ad revenues are largely dependent on the popularity of the game, and in order to have non-negligible revenues, the user-base must very important. For startup without previous blockbuster, the revenue is likely to be very low. On another hand, revenue is likely to be significant after the release of popular game.

Finally, the freemium model is considered today as the best option by almost all developers around the world. Recent blockbusters such as Candy Crush have demonstrated the opportunities that such models offer. Unfortunately, as of today, this model is very hard to implement in Africa due to the payment issues (see details in the Challenges section). Some attempts are being made to simulate in-app payments through premium SMS, but various barriers such as prices, and unpredictability of costs are not making this option viable.

As a side note, it is important to note that such models are not exclusive, and usually a game integrates multiple sources of revenues. Freemium model is usually coupled with advertisement model.

The overall conclusion that most interviewees draw is that generating revenue today from games in Africa is extremely difficult and this is largely due to payment challenges.

Startup business models

The take-off of the gaming sector is largely related to the potential sustainability of games studio. For this reason, we have investigated this area and explored with various interviewees the source of revenues for these companies.

Almost all the interviewees provided the same feedback on both the business models, and the challenges. In terms of business models, all game studios have the same approach based on mixing activities that are able to pay the development team, and the development of games that will bring revenue later, after the investment of development time.

Concerning the activities that generate revenue for the development team, there are clearly three areas:

- Serious Game development
- Outsourcing from International Studios
- Other non-game related development and consulting

Based on the interviews, it appears that across countries, the first category is the one that was cited by all the developers. There is a growing interest by all types of donors (public donors, private donors, charities, international organization etc.) to invest and explore the potential of serious games. They are therefore launching in across countries various development and evaluations. Because the market is usually local (public donors are using their local arm in a given country to recruit local developers), this option seems the easiest to access for game developers. The third category is also popular, as the so-called "mobile for development" sector is developing quickly, and there is a growing demand for local developers in various sectors such as health, agriculture, etc.

Concerning the second category, outsourcing, this is not yet very developed. With the exception of South-Africa, where a few studios are exclusively working on such model, in

in all other countries covered by the study, we only find a couple of examples of such outsourcing process. It is interesting to note that in absolutely all the cases we covered, the outsourcing process is just a matter of finding cheaper developers for the development of games targeting northern market. We didn't find any example of international game studios using this process to develop locally-adapted games that would target the African market. However, one interviewee highlighted the fact that it may happen in the near future, and this is a risk for the local startups.

In terms of challenges, all interviews highlighted a series of factors that are impacting the development of the industry. At the top of these challenges, the specificities of the game development are very problematic. Indeed, Game development requires a multi-disciplinary team with various skills from software development, to graphic design to user interaction design. The setup of such a team is not only difficult (see the challenge section about capacities), but also costly. On the other hand, generating revenue from games is extremely difficult. These two factors are creating uncertainty for investors that are not aware of the domain, despite all the available business forecast related to the gaming industry. The next section details all these challenges.

VI. Challenges

The section summarizes the key challenges that game developers and game studios are facing today that explain the reason for the relative non-take-off of the sector despite the opportunities, the size of the market and the interests of all stakeholders. In the interviews, we have identified 4 major challenges that are described below:

- Payments
- Visibility & Discoverability of games
- Access to finance
- Access to capacity

Payments

Payment is by far the primary issue that was mentioned by all interviewees. This is a major problem that is affecting both game developers that cannot be easily paid through e.g. normal store process, and gamers that cannot pay for download or in-app features due to either African credit card not accepted, or, in most cases, because gamers do not have credit card. This is currently preventing the generation of any revenue from the local market, except through advertisements.

Some studios are trying to find workarounds and have been trying to implement in-app payment through the use of premium SMS: a gamer sends a message to a premium rate number and receives a code to enter in the application. While not perfect in terms of user experience, it is today the only solution that a game company can implement at the individual level. Unfortunately, due to various barriers such as the cost for end-users, its unpredictability and various stories of abuse of premium rates by some service providers does not make this option appealing or really viable in its current form.

According to most interviewees, mobile money for paying developers and as a way for gamers to buy apps or in-app features is a promising option, particularly in countries

where it is popular like Kenya and Senegal. In South-Africa this is not a popular service. Other potential solutions mentioned by interviewees include the use of a similar model as pre-paid scratch card to pay for app or in-app payments.

The potential of carrier-billing model is unclear. During interviews, most of the interviewees were not aware of such models in their respective countries. They were doubtful that people are aware of this method of payments and ready to use it. On the contrary, during the desk research, we found a recent study (see figure 10 below) from Infodev that shows that carrier-billing would be the preferred method of payments from users in Ghana and Nigeria. We were not able to find any specific studies about the social dimension (user acceptance) of carrier billing. This point should be explored further.

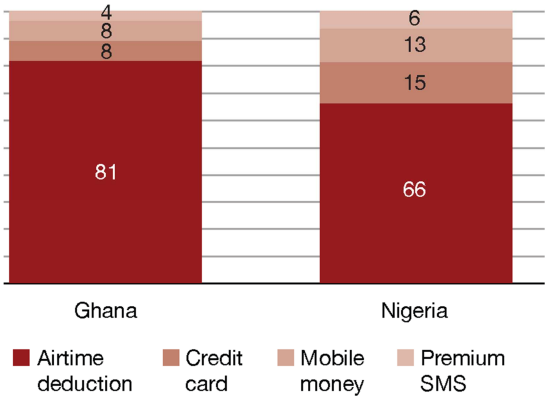


Figure 10: Preferred payment methods for App download and in-App payments by feature and smartphone users (Source Infodev Study[16])

Finally, it is interesting to note that the payment issue is not limited to game but to the whole app ecosystem. However, the specificity of games in the in-app payments, that is rarely found in other type of apps.

Visibility & discoverability

The second major issue reported by almost all interviewees is related to visibility and discoverability of games. As of today, the stores are too big to offer proper visibility. Obviously when one game has been popular it is easier to link next series of games. But for new comers, it is a major issue. Word of mouth and social networks are the current mechanisms used by game developers, and this reaches only a small audience.

The best solution to address this problem seems a bit controversial. For lots of interviewees, the availability of an operator portal where local games would be highlighted is a perfect answer to this challenge. However, for some developers who have past experience with mobile operators, they are skeptical about such a solution, particularly if it comes at a price similar to premium rate SMS service. Therefore, such a solution might be interesting, only if the cost is seen as affordable to developers, and come as a package that include payment methods.

[16] http://www.infodev.org/infodev-files/mobile_apps_at_the_base_of_the_pyramid_summary_report.pdf

Finally, as a side note, a couple of developers mentioned the need for game studios and individual developers to understand reputation on portals. The publication of un-finished work, or non-professional-level content, if not correctly labeled, can have a terrible effect on the developer reputation, impacting heavily the future uptake of new games from the same developer or company. Building a good reputation is hard, but recovering from a bad reputation is almost impossible. Bad reputation includes also poor management of advertisements.

Access to finance

Access to finance is the third major issue we identified that the game studios are facing. While the number of investors in tech startups and particularly mobile startups has been increasing in the last few years, the interest for game seems relatively low. The game business models and investment has some specific characteristics, and require the investors to understand the model. For instance there is a need for a bigger team than usual mobile applications, and the return on investments takes longer as the game needs to be fully developed before it can generate revenue.

The second issue, as discussed earlier in this section, is payments and revenue generation. It seems relatively clear that game studios targeting international markets are taking a harder route than those targeting the local market. It is now very difficult to become visible and known in e.g. Google Play given the investment of major studios in marketing and alike. On the other hand, given the lack of payment solutions, generating revenue in the local market is very difficult. This challenge is obviously a major barrier for investor. However, it is interesting to note there are examples of successful investment. E.g. Savannah Fund[17] successfully invested in Kola Studio in Uganda. In the same way, Kuluya[18] in Nigeria is now evaluated at 2MUSD and was launched through a public investment (according to one interviewee). Those examples can serve as shining examples for future investors in the domain.

Access to capacities

Finally, the lack of dedicated skills was also mentioned by most of the interviewees. Apart from South Africa, where game is now considered as a real industry and there are dedicated courses and curriculum on game developments at various universities, in all other countries, developers highlighted the lack of specific skills, in particular:

- Graphic Design: there are very few graphic designers, and in most cases they haven't received any training on the specificities of game graphical design
- User Interaction and User Experience design: these skills are globally not present, and there is no training on the matter, not only in for game design, but also for other mobile applications

The lack or shortage of these skills is impacting the ability of game studios to build the required multi-disciplinary team.

In some countries, e.g. Kenya or Senegal, some interviews mentioned that some international companies have organized specific training on e.g. specific 3D engine, in

[17] http://savannah.vc/
[18] http://kuluya.com/corporate/what.html

order to outsource some developments to the trained people. This model is interesting to explore further as it is a win-win situation where local game developers acquire skills and get paid development work from international companies, and these companies are able to develop their product at lower costs compared to the cost of game developers in western countries.

VII. Conclusion

This study highlighted a series of important facts related to the gaming industry in Sub-Saharan Africa. Concerning the community, there is a growing interest in all countries we covered for game development. All the technology hubs are reporting sessions, and groups of developers discussing game developments. Major competitions in the continents are now receiving submissions of games, and some of them are even winning these competitions. In the same way, there is a growing interest and investments from the donor community and the development sector to explore the role of serious games to address social issues and support formal education. The market is also evolving. Teenagers and young adults all over the regions play games. In the same way, there is no doubt about Android being the primary platform. Feature phones are hard to address due to interoperability issues, and the target group is starting to be equipped with smartphones due to the price war on devices. The opportunities are significant.

However, except in Nigeria and South-Africa, this is still a notion of promise and no real take-off of the sector. The major challenge at this point in time is that the offer is not meeting the demand. There are a few evidences that show that gamers are particularly interested in games that are related to their culture, and that are heavily localized. However, except in the case of serious games, lots of game developers and game studios are targeting the international market and not the local ones. The primary reason for this is the lack of payment options for the local market, making the collection (for developers) and generation (for gamers) of revenue very difficult. The second major reason is the visibility of the local games on the local market. The lack of in-country appropriate distribution and advertisement platforms make the emergence of local games very difficult.

Given the shape of the international market with firms (e.g. King[19]) that have huge team and millions of investment, it is impossible for small startups to really compete at the international level. The sustainability of game studios that target the international market is questionable. It seems far more promising and far easier to target the local market. Given the context, (presence of demand, presence of potential offer, interest and excitement from all stakeholders), it is likely that a series of focused interventions focusing on the core barriers can boost the sector and provide a win-win situation for game developers, mobile operators, gamers and development organizations interested in investigating the potential on games.

[19] http://company.king.com/

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